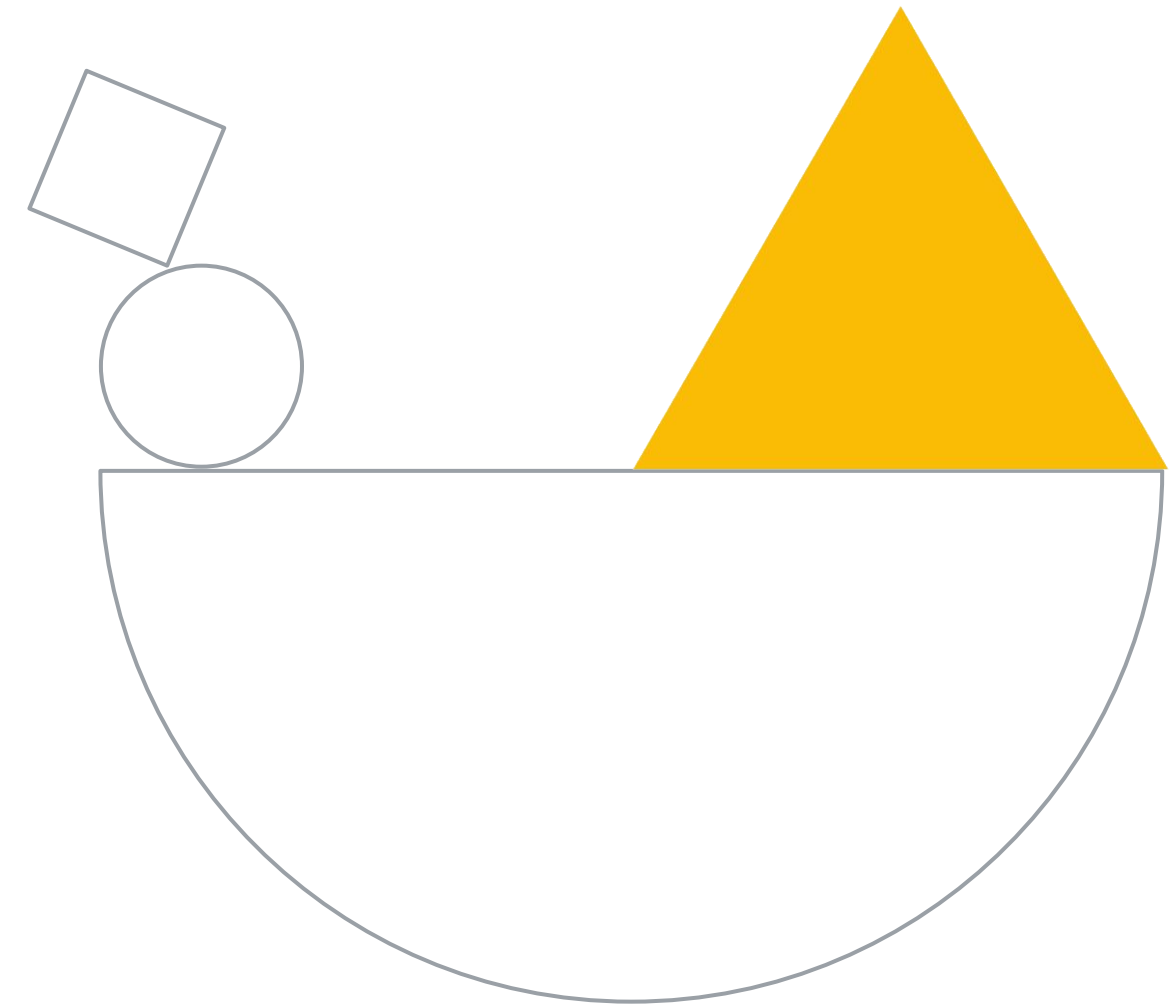




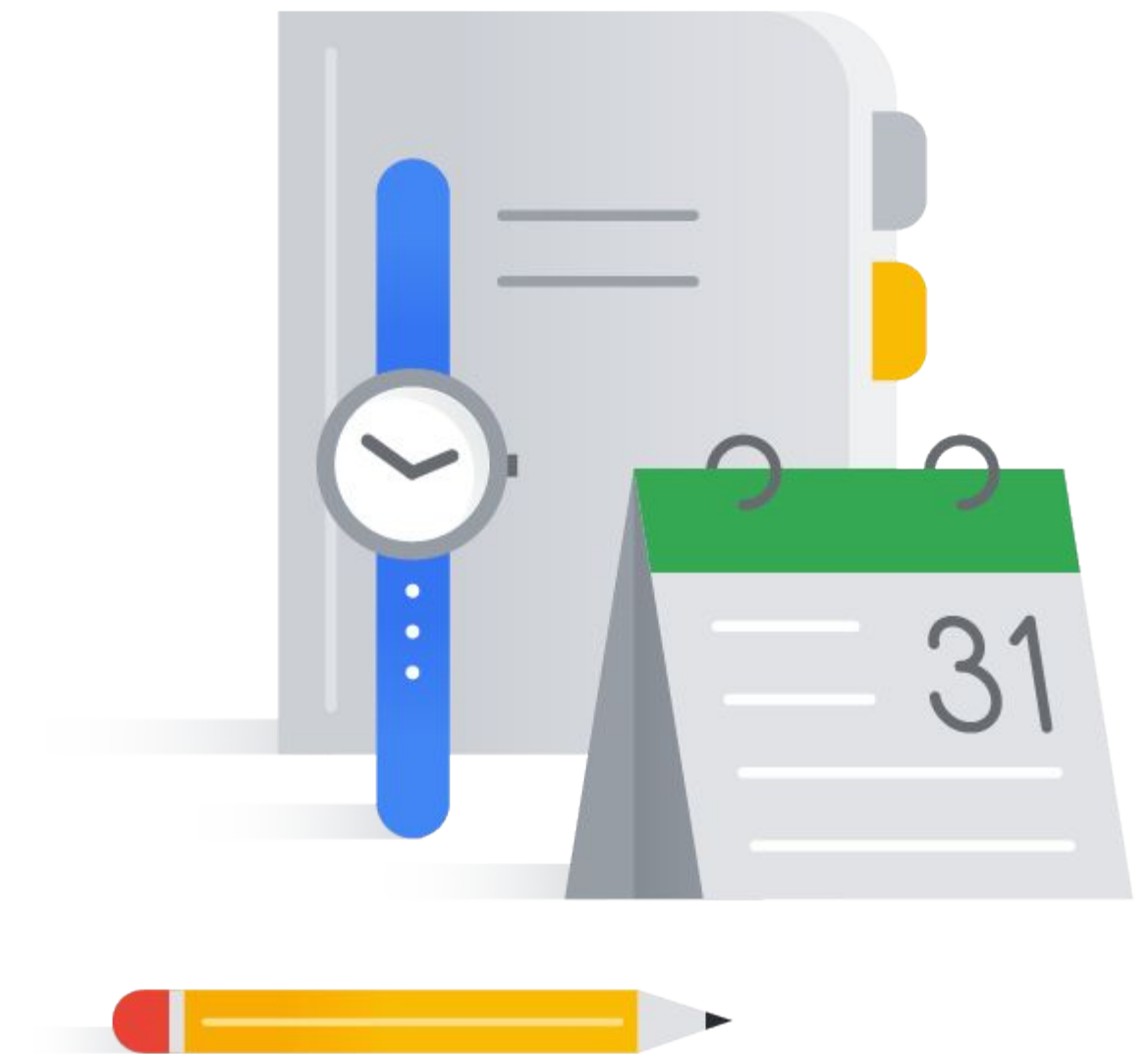
Infrastructure Automation



Agenda

- 01 Terraform
Lab: Automating the Deployment of Infrastructure
Using Terraform

- 02 Google Cloud Marketplace

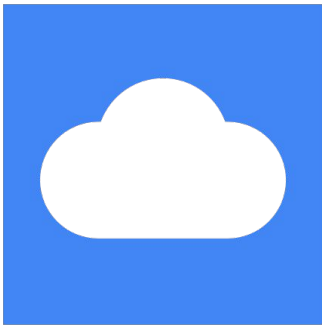




Terraform

Google Cloud console, Cloud SDK, and Cloud Shell

Google Cloud console



console.cloud.google.com



Cloud Shell

Google Cloud SDK

☐	Name ^	Zone	Internal IP	External IP	Connect	
☐	✔ nginxstack-1	us-central1-f	10.128.0.3 (nic0)	35.238.84.245	SSH ▾	⋮
☐	✔ nginxstack-2	us-central1-f	10.128.0.4 (nic0)	35.225.177.18	SSH ▾	⋮
☐	✔ nginxstack-3	us-central1-f	10.128.0.2 (nic0)	35.239.250.238	SSH ▾	⋮

\$ gcloud compute instances list

NAME	ZONE	INTERNAL_IP	EXTERNAL_IP
nginxstack-1	us-central1-f	10.128.0.3	35.238.84.245
nginxstack-2	us-central1-f	10.128.0.4	35.225.177.18
nginxstack-3	us-central1-f	10.128.0.2	35.239.250.238

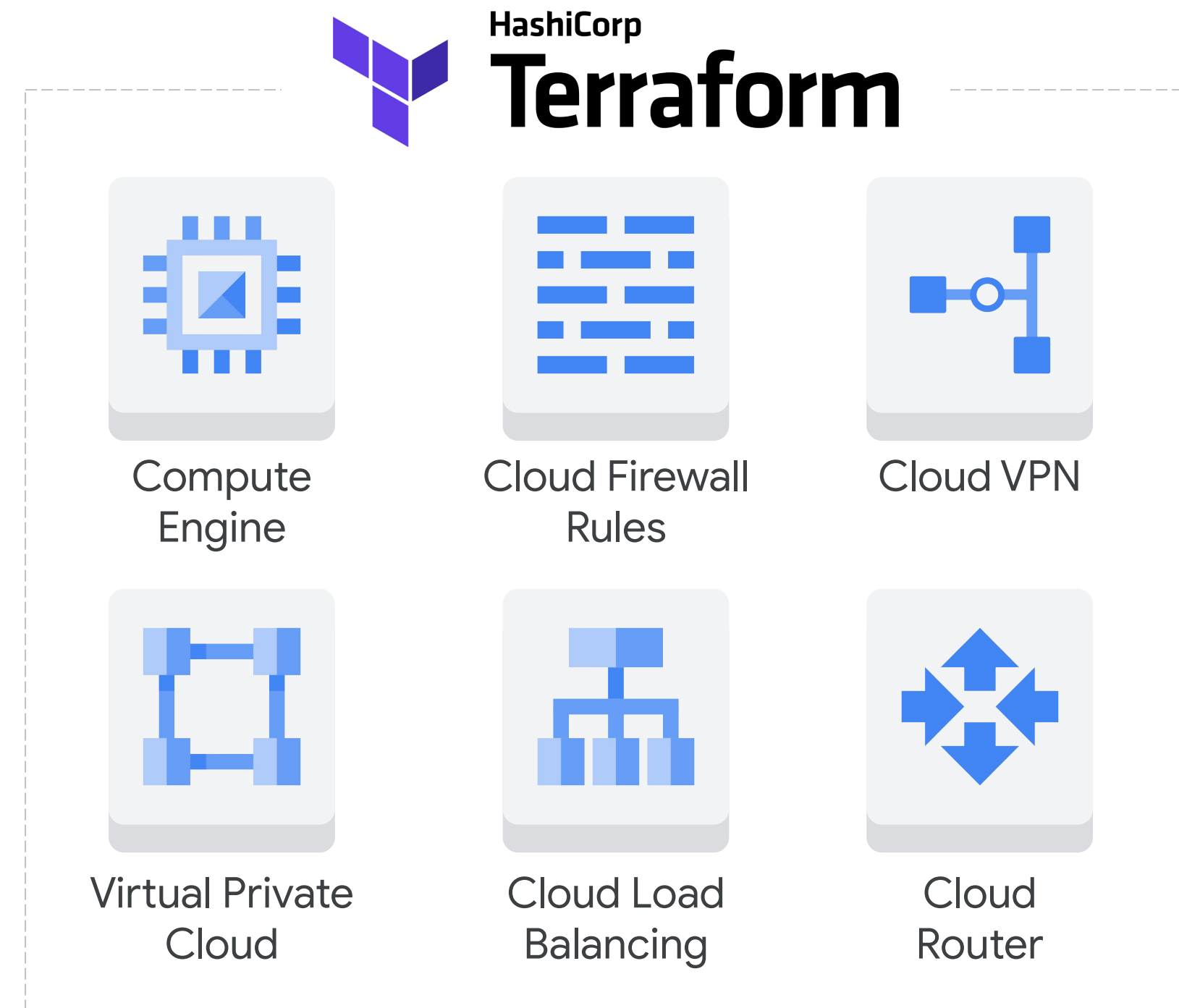
Infrastructure as code (IaC) allows quick provisioning and removal of infrastructures

- Build an infrastructure when needed.
- Destroy the infrastructure when not in use.
- Create identical infrastructures for dev, test, and prod.
- Can be part of a CI/CD pipeline.
- Templates are the building blocks for disaster recovery procedures.
- Manage resource dependencies and complexity.
- Google Cloud supports many IaC tools.



Terraform is an infrastructure automation tool

- Repeatable deployment process
- Declarative language
- Focus on the application
- Parallel deployment
- Template-driven



Terraform language

- Terraform language is the interface to declare resources.
- Resources are infrastructure objects.
- The configuration file guides the management of the resource.

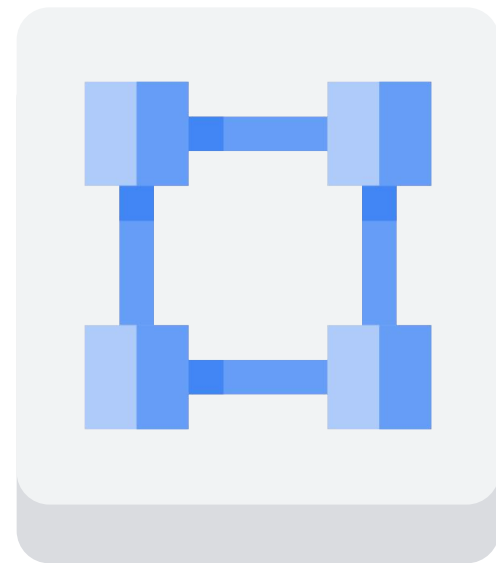
```
resource "google_compute_network" "default" {  
  name = "${var.network_name}"  
  auto_create_subnetworks = false  
}  
  
<BLOCK TYPE> "<BLOCK LABEL>" "<BLOCK LABEL>" {  
  # Block body  
  <IDENTIFIER> = <EXPRESSION> # Argument  
}
```

Terraform can be used on multiple public and private clouds

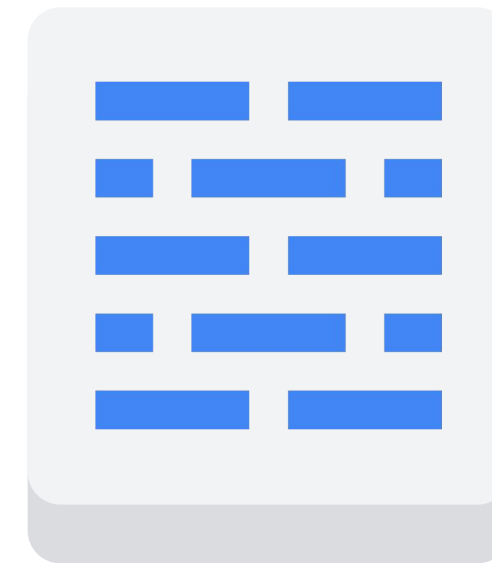
- Considered a first-class tool in Google Cloud
- Already installed in Cloud Shell

```
provider "google" {  
    region      = "us-central1"  
}  
  
resource "google_compute_instance" {  
    name          = "instance name"  
    machine_type = "n1-standard-1"  
    zone          = "us-central1-f"  
  
    disk {  
        image = "image to build instance"  
    }  
}  
  
output "instance_ip" {  
    value = "${google_compute.ip_address}"  
}
```

Example: Auto mode network with HTTP firewall rule



Auto mode
network



HTTP firewall
rule

Example: Auto mode network with HTTP firewall rule

```
main.tf
provider "google" {
}
```

Example: Auto mode network with HTTP firewall rule

```
main.tf
provider "google" {
}
# [START vpc_auto_create]
resource "google_compute_network" "vpc_network" {
  project          = var.project_id
  name             = "my-auto-mode-network"
  auto_create_subnetworks = true
  mtu              = 1460
}
```

Example: Auto mode network with HTTP firewall rule

```
main.tf
provider "google" {
}
.
.
.

# [START vpc_firewall_create]
resource "google_compute_firewall" "rules" {
  project      = var.project_id
  name         = "my-firewall-rule"
  network      = "vpc_network"
  allow {
    protocol = "tcp"
    ports    = ["80", "8080"]
  }
}
```

Deploying infrastructure with Terraform

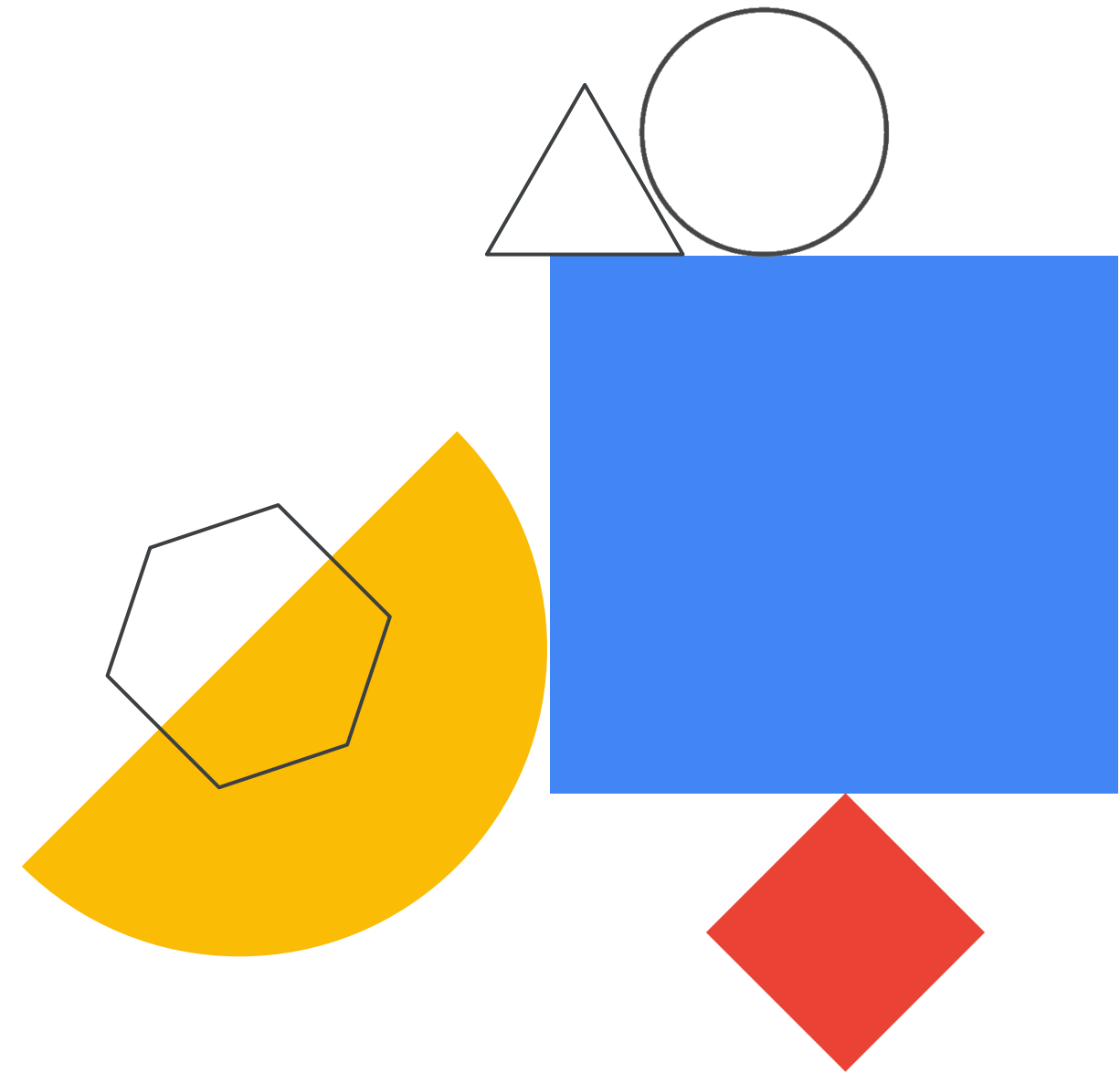
```
terraform init
```

```
terraform plan
```

```
terraform apply
```

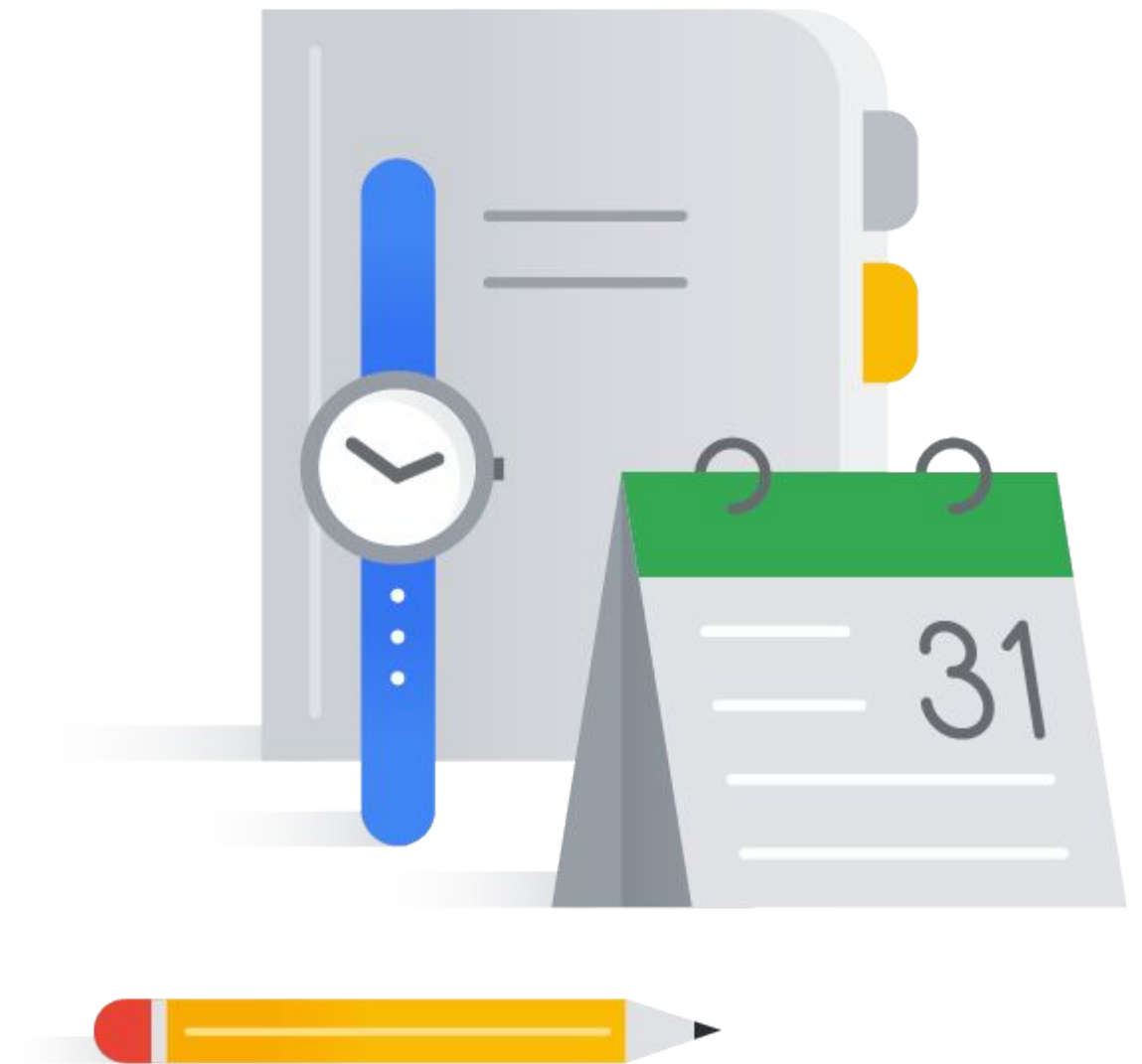
Lab intro

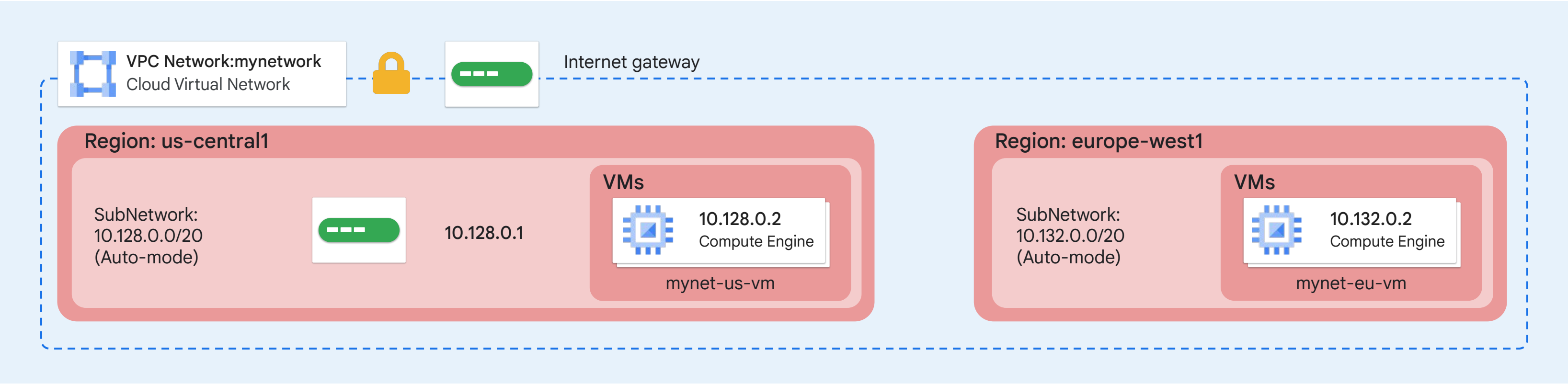
Automating the Deployment of
Infrastructure Using Terraform



Lab objectives

- 01 Create a configuration for an auto mode network.
- 02 Create a configuration for a firewall rule.
- 03 Create a template for VM instances.
- 04 Create and deploy a configuration.
- 05 Verify the deployment of a configuration.





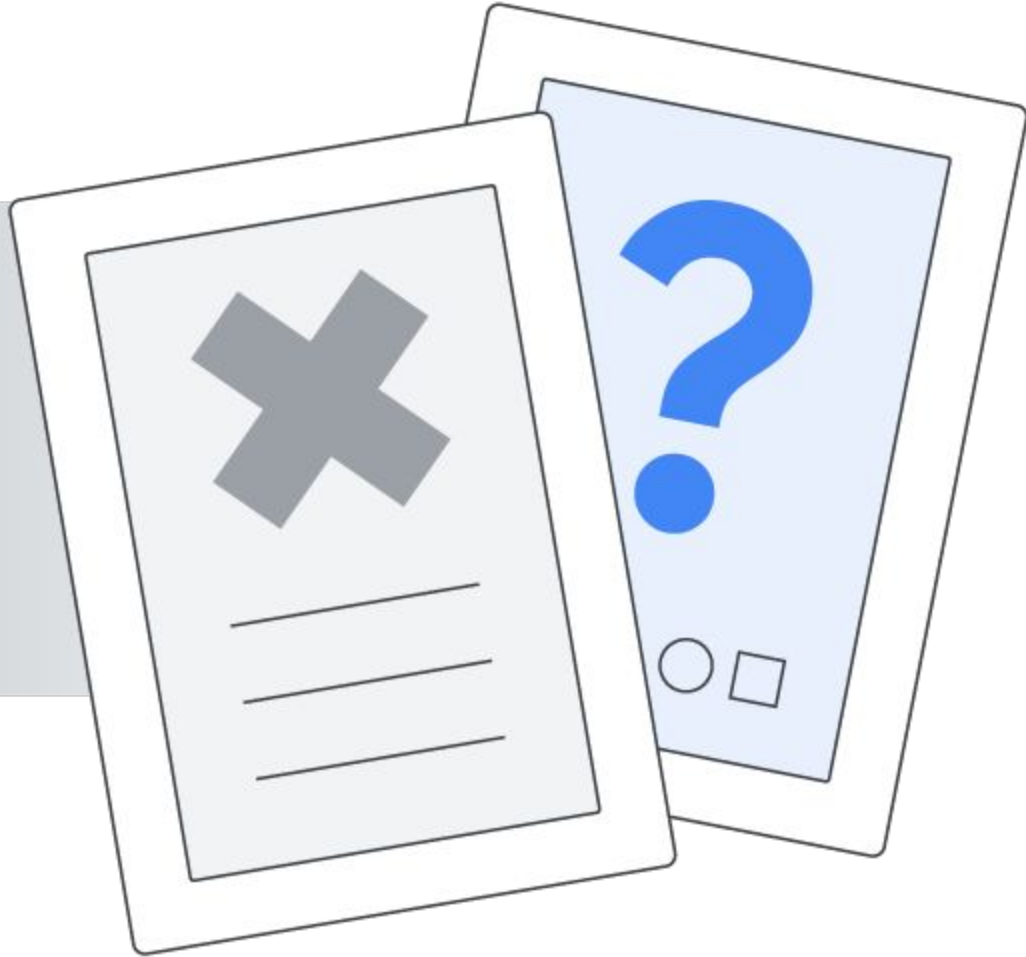


Google Cloud Marketplace

Google Cloud Marketplace

- Deploy production-grade solutions
- Single bill for Google Cloud and third-party services
- Manage solutions using Terraform
- Notifications when a security update is available
- Direct access to partner support

Quiz



Question #1

Question

What's the benefit of writing templates for your Terraform configuration?

- A. Allows you to run configuration management software.
- B. Allows you to hardcode properties for your resources.
- C. Allows you to abstract part of your configuration into individual building blocks that you can reuse.

Question #1

Answer

What's the benefit of writing templates for your Terraform configuration?

- A. Allows you to run configuration management software.
- B. Allows you to hardcode properties for your resources.
- C. Allows you to abstract part of your configuration into individual building blocks that you can reuse.



Question #2

Question

What does Cloud Marketplace offer?

- A. A centralized billing platform for all Google Cloud services and applications
- B. Production-grade solutions from third-party vendors who have already created their own deployment configurations based on Terraform
- C. A platform for trading VM instances

Question #2

Answer

What does Cloud Marketplace offer?

- A. A centralized billing platform for all Google Cloud services and applications
- B. Production-grade solutions from third-party vendors who have already created their own deployment configurations based on Terraform
- C. A platform for trading VM instances



Review: Infrastructure Automation

